

Advancing sustainable safety National road safety outlook for The Netherlands for 2005–2020

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Abstract

Although road safety has improved enormously over time, and The Netherlands is one of the safest countries in the world, the current annual number of road casualties is still considered as unacceptable. To support next steps the sustainable safety vision was launched in the early 1990s. The idea was to make the Dutch road traffic system inherently safe. Sustainable safety was the guide in the approach of improving road safety in The Netherlands the last two decades. Because implemented sustainable safety measures reduced the number of casualties, the vision can be considered as successful. More than a decade later, SWOV Institute for Road Safety Research judged the time ready for evaluating and updating the original Sustainable Safety vision. The advanced sustainable safety vision is meant to inspire the road safety community and all tiers of government in the way they may improve road safety in the coming 15 to 20 years. This paper presents the (theoretical) core of the advanced sustainable safety vision: the aims of preventing crashes, and where this is not possible, to reduce the chance of severe injury to (almost) zero; the proactive approach to reach this, with ‘man as the measure of all things’ as a starting point, in a balanced combination of the elements ‘road’, ‘vehicle’, and ‘man’. In the sustainable safety vision, five principles are central now: *functionality* of roads, *homogeneity* of masses and/or speed and direction, *predictability* of road course and road user behaviour by a recognizable road design, *forgiveness* of the environment and of road users, and *state awareness* by the road user. The vision has been translated into numerous ideas for practical proposals. Special attention is paid to implementation of these proposals given the political and governmental context of The Netherlands.

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1. Introduction

In The Netherlands, a little less than one thousand road users are killed and many tens of thousands are injured each year. Compared to other countries, The Netherlands performs very well, and it is one of the safest countries in the world. Currently The Netherlands tops the world in having the lowest number of fatalities per

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capita. Dutch road safety policy is often identified as good practice, and the Sustainable Safety vision as leading practice (Peden et al., 2004). Dutch road safety performance commands respect.

At the same time, every year, we have to deplore the occurrence of so many road traffic casualties. This represents enormous societal loss. It was calculated that this cost Dutch society 12.3 billion Euros in 2003, which amounts to 2.6% of the Gross National Product, including the costs of injuries and material damage caused by road crashes. These costs also comprise intangible costs that are calculated for loss of quality of life for victims and their surviving relatives (SWOV, 2005).

Every year, the disaster of so many casualties in road crashes is not perceived as a disaster, and does not get the response that is commensurate with a disaster. One crash with one thousand people killed is a disaster; one thousand deaths in one thousand crashes are as many individual tragedies. The average citizen seems to shrug it off as if all these anonymous deaths in traffic crashes are just part of life. The risk of being killed in a road crash seems too abstract a concept to be worried about. However, it is a different story when a fatally injured party is a neighbour, a colleague, a good friend or your own child. Then there is great dismay about how this could possibly have happened, and questions arise as to how it could have been prevented. It is not surprising that the Dutch consider road safety to be of great personal, societal and political importance (Voorlichtingsraad, 2005). But what are the next steps?

The current size of the road safety problem in The Netherlands is characterised as unacceptable, and we strive for further reduction in the number of casualties. “Permanent road safety improvement” is not of much significance, and is more a signal that the subject is not forgotten. Formulating a task is one step further, and shows more ambition. Working with quantified targets has been commonplace in the Netherlands for decades. The level of ambition (a reduction in the number of road fatalities by 25% in ten years time) is not out of the ordinary when compared with other countries. The ambition formulated by the European Commission – halving the number of road fatalities in ten years time – is highly ambitious (European Commission, 2001), but has resulted, without any doubt, in the subject being on the agenda in Europe. In several Member States it has led to renewed attention and continuing efforts.

The Dutch *Mobility Policy Document* states that, while absolute safety and total exclusion of risk does not exist (Ministry of Transport, 2004), the number of casualties can, without any doubt, be further reduced. There is no lack of ideas, but the question is: at what cost? To this end, SWOV has proposed using the criterion of “avoidable crashes” (Wegman, 2000). “Avoidable” in this context means that we know what to do in order to prevent crashes and that it is cost-beneficial in societal terms to do this. In other words: the benefits exceed the costs. Seen from considerations of effectiveness and efficiency, we later added “and fitting within the sustainable safety vision”.

Advancing sustainable safety (Wegman and Aarts, 2006) is the follow-up to “Naar een duurzaam veilig wegverkeer” (Towards Sustainable Road Traffic Safety) (Koornstra et al., 1992). In this advanced version adaptations were made where necessary, based on what we have learned from our first steps towards sustainably safe road traffic. The vision has also been updated in line with new insights and developments.

We chose a broader perspective for this book than in 1992. This broader perspective is justified, because we have been able to evaluate the results of our efforts to date. Moreover, there was high demand from practitioners to develop sustainable safety for specific problem areas or problem groups. Finally, this perspective offers the opportunity to “position” the vision again, and to eliminate any misunderstandings. In this manner, we want to provide a new stimulus for the further implementation of sustainable safety.

2. Road safety: past, present, future

The number of fatalities in road traffic peaked with a record of 3264 in 1972. This amounted to about 9 deaths every day. After 1972, the increase in deaths reversed, despite ever increasing mobility, and a downward trend has been maintained. The number of deaths first decreased sharply (except during the period 1975–1977), but from the mid 1980s onwards this trend became somewhat less pronounced (see Fig. 1). In 2006, 811 fatalities in road crashes were reported, four times less than in 1972.

In general, road safety has improved enormously over time, and the Netherlands is one of the safest countries in the world (Fig. 2). The rate of improvement in The Netherlands has also been high in the past decades. This can partly be attributed to a learning society, which has grown used to modern, fast traffic. In addition,

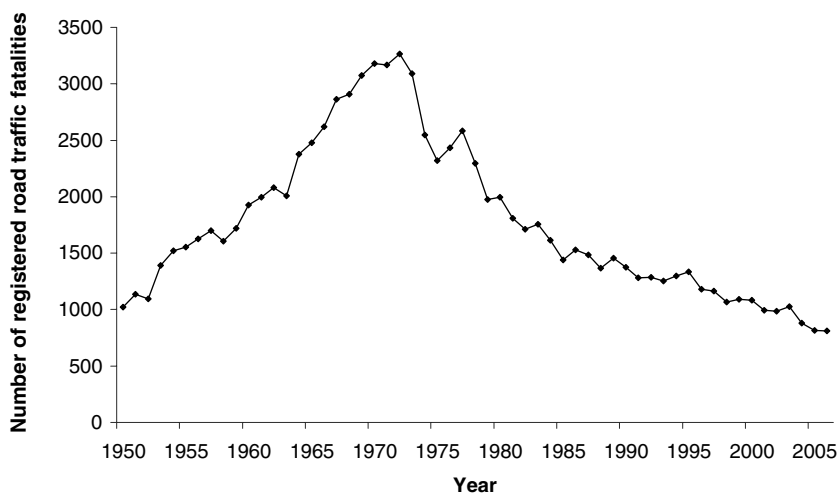


Fig. 1. The number of registered road fatalities per year in The Netherlands in the period 1950–2004. Source: AVV Transport Research Centre.

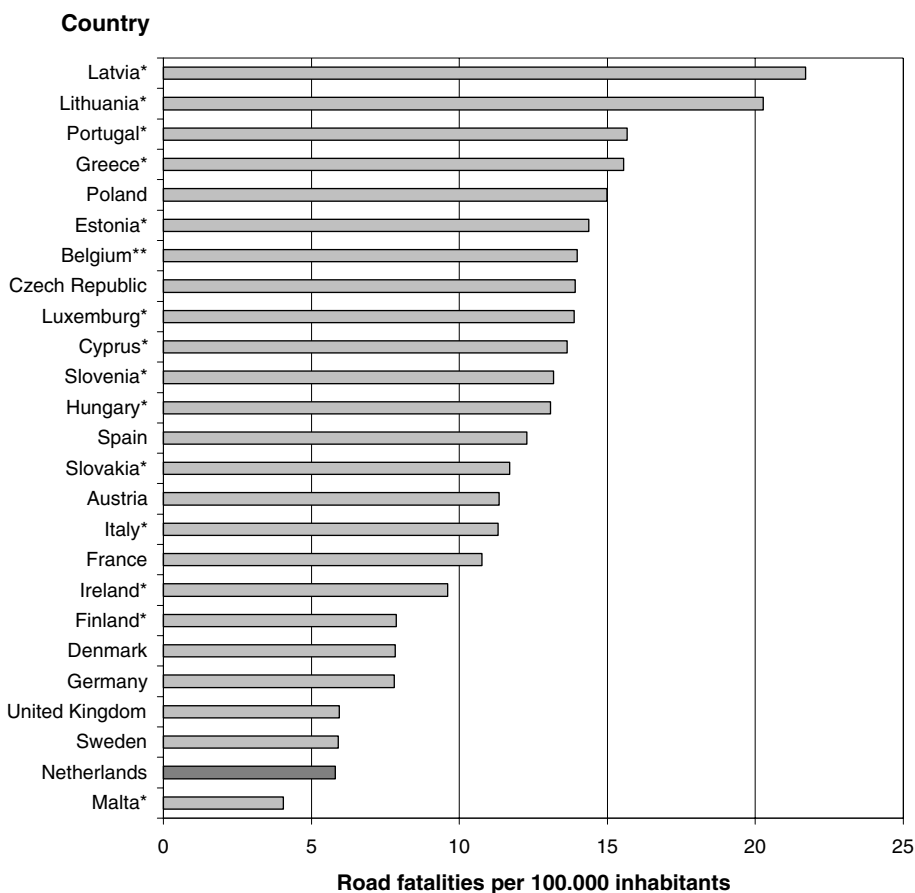


Fig. 2. Number of road fatalities per 100,000 inhabitants for the current 25 EU Member States averaged over 2002–2004 (* = 2001–2003, ** = 2000–2002). Sources: IRTAD; CARE; Eurostat.

infrastructural adaptation has taken place (such as the construction of relatively safe motorways), passive safety in vehicles has been improved, and there is more safety legislation and enforcement which takes account of factors which increase risk and reduce injury (such as alcohol consumption in traffic, and mandatory crash helmet and seat belt use, respectively). These measures have also contributed to reductions in the number of traffic fatalities and injuries, despite increased mobility. The SUNflower research has made these possible explanations plausible (Koornstra et al., 2002) and is supported by other scientific literature (Elvik and Vaa, 2004). But, as yet, we do not have a totally conclusive explanation for these improvements.

2.1. *Fundamental risk factors in traffic*

Fundamental risks which are inherent to road traffic are the basis of the lack of safety in current road traffic. These are factors such as speed, mass and vulnerability. With fundamental factors we do not mean those factors that form the foundation of the process towards a safer system (see TRIPOD-model; e.g. van der Schrier et al., 1998).

2.1.1. *Speed*

Speed is not only a given fact in traffic; it is also a fundamental *risk* factor. Firstly, speed is related to *crash risk* (for an overview, see Aarts and van Schagen, 2006). From several studies of the relationship between speed and crash risk, we can conclude that higher absolute speeds of individual vehicles are related to an exponential increase in risk (Kloeden et al., 1997, 2001). Speed differences are also linked with increases in crash risk (e.g. Solomon, 1964). Recent research however, has not proven that vehicles travelling at lower speeds than the traffic flow have a higher risk than vehicles that go with the flow (e.g. Kloeden et al., 1997, 2001). At the same time, vehicles going faster than the traffic flow have an increased risk. Speed variance at the level of road section is also linked to increased crash risk (e.g. Taylor et al., 2000). Secondly, speed is related to *crash severity*. This is based on the kinetic energy (of which speed is an important indicator), which is converted into other energy forms during a crash, causing damage in the process. Injury risk is also determined by speed level, the relative directions of crash partners, their mass differences and protection level, and biomechanical laws.

2.1.2. *Mass and protection*

While in motion, the total mass of a vehicle combined with its speed produces kinetic energy, which is converted into other energy forms during a crash and can cause material and/or bodily damage. In a crash between two incompatible parties, the lighter party is at a disadvantage because this party absorbs more kinetic energy and the vehicle generally offers less protection to its occupants than a heavier vehicle. Mass differences between colliding vehicles can amount to more than a factor of 300 (a pedestrian weighing 60 kg versus a heavy goods vehicle weighing 20,000 kg). Furthermore, in view of their stiffness and structure, heavier vehicle types generally offer better protection to their occupants in the event of a crash. For occupants of vehicles with a high mass, injury risk is much lower than that of the lighter crash party. The injury factor increases to 1.4 if the crash opponent weighs 1000 kg, and increases to factor 1.8 if the crash opponent weighs 1500 kg (Elvik and Vaa, 2004).

2.2. *Risk increasing factors from the road users' side*

Besides these fundamental factors we can distinguish a few risk increasing factors, and research results teach us about the relevance of these different factors.

- *Lack of driving experience.* The effect of driving experience on crash risk is strongly linked with age effects. Research into the influence of driving experience on crash risk, indicates that about 60% of the crash risk of novice drivers can be explained by lack of driving experience (e.g. Sagberg, 1998). From this research it can also be concluded that the increased crash risk of novice drivers (a factor of 2.5 relative to drivers with more than five years of experience) decreases rapidly within the first year after passing a driving test. Since driving

experience is strongly correlated with age, and since both factors are associated with specific characteristics which increase risk, it is difficult to separate age and experience.

- *Psycho-active substances: alcohol and drugs.* Alcohol is one of the most important factors which increase risk in traffic. Crash risk increases exponentially with increased blood alcohol content (BAC). Recent research into the crash risk of road users under the influence of psycho-active substances revealed that the risk is about a factor of 25 higher with the combined use of drugs. Also there is cumulative road crash fatality risk when combined with the use of alcohol and drugs (BESEDIM et al., 1997).
- *Illnesses and ailments.* Visual limitations or ailments are generally associated with a very small increase in crash risk (reduced useful field of view and glare sensitivity). Decreased hearing only results in a slightly increased risk. The same is the case for neurological disorders, for people with Alzheimer's disease and other psychiatric disorders, such as cognitive disorders and depression (Vlakveld et al., 2005).
- *Emotion and aggression.* It seems to be difficult to draw conclusions about the relationship between aggression and road safety. The literature (e.g. Deffenbacher et al., 2003; Mesken et al., 2002; Stradling et al., 1998) leaves the impression that there is a coherent behavioural pattern of a combination of various aggressive and/or risky behaviour types that result in a dangerous driving style. However, for the time being it is not possible to quantify the risk associated with this risk factor.
- *Fatigue.* Fatigue is most probably a much more frequently occurring factor in increasing risk than data from police reports shows. Research shows that people suffering from a sleep disorder or an acute lack of sleep, have a 3–8 times higher risk of injury crash involvement (Connor et al., 2002).
- *Distraction.* Like fatigue, distraction is probably a much more frequent crash cause than reported police data show. Currently, one of the more common sources of distraction is use of the mobile phone while driving. The permitted hands-free option does not reduce the effect of distraction either (e.g. Patten et al., 2004). The most well-known and best research into the risk of using a mobile phone while driving indicates an increase in risk by a factor of 4 relative to non-users (Redelmeier and Tibshirani, 1997; McEvoy et al., 2005). Other studies show a similar risk increase (for an overview, see Dragutinovic, 2005). Other activities such as operating route-navigation systems, tuning CD-players and radios etc. can also be a source of distraction, as can activity such as eating, drinking, smoking and talking with passengers (see Young et al., 2003).

2.3. Increased understanding of road traffic safety

Through the years, we have gained a better insight in factors which influence road safety. Much can be understood about road safety from the fundamental risk factors: speed, mass and vulnerability. Research results from the past teach us this. They also identify where the opportunities are for improvement. Users of motorized two-wheelers have the highest fatality and injury risk in road traffic, which can largely be explained by a combination of high speed with the relatively low mass of the vehicle in conflict with other motorized traffic, as well as poor crash protection. On top of that, mopeds are popular with young people who have yet to obtain a driving licence. This group already has a relatively high risk in traffic because of age-specific characteristics and needs, and lack of experience.

Safety on roads can also to a large extent be explained by a combination of fundamental risk factors. For example, serious crashes outside urban areas, and particularly on rural 80 km/h roads, are dominated by single vehicle conflicts along sections of road. These are usually the result of inappropriate speeds, possibly in combination with other factors which increase risk, such as alcohol consumption, distraction and/or fatigue, and the fact that many roads are not “forgiving”, results in errors being punished with (severe) outcomes. Intelligent Transport Systems that keep speeds within limits or which monitor the driver's state, could reduce risk here, but the road and roadside could be designed in such a way that errors are not punished with serious outcomes. On urban roads, transverse conflicts predominate. On 50 km/h roads, in particular, where most people are killed in urban areas, mass differentials and the vulnerability of road users are important factors, combined with comparatively high speed, and the vulnerability of vehicles in transverse conflicts. In The Netherlands the risk of being involved in a crash is highest on urban 50 km/h and rural 80 km/h roads. Motorways are the safest roads when it comes to crash risk. This is due to a combination of road design and vehicles

allowed on this type of road. The road design is appropriate for high driving speed, both physically (separation of driving direction) and in a psychologically (predictable design) so that high speeds can be managed in relative safety.

2.4. *What will the future bring?*

So much for past and present. But what will the future have in store for us? In order to determine the appropriate strategy in a road safety vision and to propose the right measures, we need to take account of future societal changes. After all, sustainable safety has the ambition to be proactive to anticipate possible dangerous developments, tendencies and situations, instead of taking action after serious crashes have taken place. In the next section, a number of current developments relevant to traffic and road safety in The Netherlands are outlined.

- *Mobility.* The first development that is expected to have an effect on future road safety is further growth in mobility. This growth can mainly be attributed to economic and population growth (Statistics Netherlands, 2004). It is expected that this will set two developments in motion. Firstly, further growth in car mobility, especially car use for social activities (Social and Cultural Planning Office of the Netherlands, 2004 in Schoon, 2005), with which extension of the road network will not keep pace. This means that traffic will become increasingly busy, and traffic will also be distributed more evenly over time (“the off-peak hours between peak hours will fill up”) and place (“more cut-through traffic”). The exact consequences for road traffic are difficult to assess, but they will depend upon the way in which people react to ever heavier traffic. More conflict possibilities will occur, for example because the relatively dangerous secondary road network will be used to relieve the main road network. At the same time, when so much traffic has to be accommodated, the increased intensity will also result in lower speeds, with less likelihood of serious crashes. Modal shift may also occur, concurrent with mobility growth. A second development is an increase in mileage by heavy goods vehicles and vans. This is also related to expected economic growth.
- *Demography.* A second development concerns demography within The Netherlands. Of particular note are the large and increasing numbers of older people, and the large numbers of young people born in the 1980s (Statistics Netherlands, 2004). Combined with the trend of increasing individualism, this is expected to result in more single households. The effects of these developments are expected to result in facilities being spread over larger areas with increasing dependence on cars (Methorst and van Raamsdonk, 2003). Another contribution is made by the life pattern of double-income families who, in combining (part-time) work, care tasks and often considerable commuting distances, will use the car more often, having previously gone on foot or taken the bicycle. The school trip, for example is even now, combined with commuting to and from work (Schoon, 2005). This trend is another source of increased car mobility, but also of increased driving experience and driving licence ownership in traffic. Since children will begin to participate in traffic at a later age, they will also need to learn to cope with traffic at a later stage (Schoon, 2005). This possibly has a negative effect on the risk of (young) cyclists and moped riders.
- *Social culture.* Developments in various cultural, societal and/or age-specific subcultures can also have their effect on road safety (Schoon, 2005). There is a trend that certain groups of young people in the lower socio-economic groups in particular (linked with certain car types and motorized two-wheelers), regard traffic as a playing field where one can indulge in risky behaviour by striving for sensation. This is also related to increased (perceived) aggression and intolerance in traffic (“we sometimes have a very short fuse in our small country”). This is possibly also related to increasingly congested traffic and resulting delays when travelling. Leaving aside some, fortunately incidental, cases of excessive aggression in traffic, the question remains as to the extent to which aggression actually leads to more crashes. Nevertheless, it seems appropriate here to keep a finger on the pulse.
- *Consumption.* The growth in prosperity, linked to the growth in disposable income, is expected not only to result in increased mobility, but also in a more rapid renewal of the car fleet. This has benefits for severe injury risk, because new cars usually have better primary and secondary safety. This will also have an influence on the increase in technological applications in motorized traffic. However, an increase in consumption

possibilities is much less positive for road safety when it comes the number of motorcyclists, increased alcohol consumption and increasing fatigue – thinking for instance of the advance of the 24-h economy.

- *Quality of life.* Increasing prosperity also results in increased importance being attached by society to the quality of life. Health, healthy lifestyles and a clean environment become important issues. These can be beneficial for road safety, not only because of decreasing acceptance of risks, but also recognition of high-quality trauma organization as a secondary effect (Amelink and Schoon, 2005; Racioppi et al., 2004). City centres which are not accessible to car traffic, and which shift mobility to the periphery of urban areas provide one example. This trend is also related to the increased densities of urban areas, where congested traffic flows offer opportunities for expanding the public transport network and reducing car mobility in residential areas (Schoon and Schreuders, 2006). In sustainable safety, there is a clear interest in taking advantage of developments in urban planning and development. However, it needs to be said that this has not yet found a firm footing. But the interest is there, and it certainly is an issue that deserves more attention.
- *Public governance.* Finally, mention must be made of trends in the governance of the public sector in relation to developments in society, and the relationship between individual and governmental responsibility. Firstly, this concerns the effects of governmental organization on road safety. In The Netherlands, this is organised in a decentralised way, but there is also the increasing influence of Europe. The decreased room for manoeuvre in government funding, the reduced staff capacity and expertise, the decreasing tendency to regulate centrally for executive organizations, together with decreased frequency of inspection of the implementation of measures, have to be compensated by the increased responsibility of citizens who, knowing more, do not like to be told what to do. In view of this, the central organization of a number of traffic and transport matters is no longer an issue, and the question arises as to what this means for road safety. In view of these developments in public governance, the high economic importance of traffic and transport, the ever increasing traffic congestion which takes up all available capacity and the fact that new consideration has to be given to accessibility, quality of life, environment, and road safety, decision making processes are becoming ever more complex. Extra effort and dedicated knowledge is required to allow full consideration of road safety in decision making. Add to all this, the fact of life of more emancipated citizens and the fact that they view road crashes as a big problem, this means a growing “market” for the societal centre ground. If we combine this conclusion with the notion that citizens’ support becomes increasingly important, then it is clear that the “road safety lobby” has an important role to play in the future. Improving road safety and realising sustainable safety will benefit from a strong road safety advocacy.

2.5. To summarize

Road safety analyses show that car mobility, in particular, has increased in the course of time, with an enormous and simultaneous improvement in safety. Much of the latter is due to large and small efforts to improve the safety of all the components of the traffic system. While mobility is expected to grow even more in future, the growth rate is likely to be lower than in the past. The growth in mobility has specific consequences for combined traffic management as the numbers of vulnerable road users such as cyclists and pedestrians increase. In The Netherlands, a large proportion of these will be elderly road users.

Motorized two-wheelers are a comparatively dangerous transport mode. This is related to the combination of relatively high speeds and a lack of physical protection. Mopeds are popular with young people, who run a higher risk of serious crash involvement due to a lack of experience and age-specific characteristics. With respect to safety on roads, rural roads with a posted speed limit of 80 km/h and urban roads and streets with a posted limit of 50 km/h deserve the greatest attention because two-third of all fatal crashes take place on these roads. Single vehicle crashes result in severely injured victims on sections of road. Injury-producing side impacts are the main problem at intersections. Large differences in mass exacerbate the injury severity of the weaker party.

The important factors which increase risk are speed, in particular, and the use of psycho-active substances (mainly alcohol and drugs). Other factors that probably cause crashes far more frequently than can be ascertained from (police) records, are issues such as fatigue and distraction. Fatigue is expected to be an increasing

problem in the future, as will distraction in an age where more and more tasks will be automated. The general crash causation picture is that anyone can make unintentional errors and while these probably comprise the lion share, violations should not to be neglected. At least, violations can substantially increase the crash risk, whether a combination of individual error, or other road user error. Preventing human error from resulting in a serious crash remains a very important issue for improving road safety. This can be reinforced by preventing violations of the limits which society has set to address known factors which increase risk.

3. The principles of sustainable safety

The objective of sustainable safety is to prevent road crashes from happening, and, where this is not feasible, to reduce the incidence of (serious) injuries whenever possible. This can be achieved by a proactive approach in which human characteristics are used as the starting point: a user-centric system approach. On the one hand these characteristics refer to human *physical vulnerability*, and to human (cognitive) *capabilities and limitations* on the other. People regularly make errors unintentionally and are not always able to perform their tasks as they should. Furthermore, people are also not always willing to comply with rules and violate them intentionally. By tailoring the environment (e.g. the road or the vehicle) to human characteristics, and by preparing the road user for traffic tasks (by training and education), we can achieve an inherently safe road traffic system.

The most important features of inherently or sustainably safe traffic are that *latent errors* in the traffic system (gaps in the system that result in human errors or traffic violations causing crashes) are, as far as possible, prevented and that *road safety depends as little as possible on individual road user decisions*. The responsibility for safe road use should not be placed solely on the shoulders of road users but also on those of who are responsible for the design and operation of the various elements of the traffic system (such as infrastructure, vehicles and education).

A set of guiding principles has been developed to achieve a sustainably safe road traffic. The old principles from the original Sustainable Safety vision have been reformulated where appropriate, and some new principles have been added. This has resulted in five sustainable safety principles as shown in Table 1.

These principles are introduced in Sections 3.1–3.3. See for more details: Wegman and Aarts (2006).

3.1. Traffic planning: functionality principle

Traffic flow organization manifests itself in many ways and with various and different objectives. As long ago as the 1960s, a functional road categorisation system was introduced (Buchanan, 1963), which formed the basis for the sustainable safety *functionality principle*. This principle starts from the premise that roads can only have a single function (*mono-functionality*) and that they must be designed and used in keeping with that function. The road function can on the one hand be “to facilitate traffic flow” (associated with “through roads”), and on the other hand “to provide access to destinations” (associated with “access roads”). In order to provide a proper transition between “giving access” and “facilitating traffic flow”, a third category of function was defined: the “distributor road”.

Table 1
The five sustainable safety principles with a short description

Sustainable safety principle	Description
<i>Functionality</i> of roads	Mono-functionality of roads as flow roads, distributor roads or access roads, in a hierarchically structured road network
<i>Homogeneity</i> of masses and/or speed and direction	Equity in speed, direction, and masses at medium and high speeds
<i>Forgivingness</i> of the environment and of road users	Injury limitation through a forgiving road environment and anticipation of road user behaviour
<i>Predictability</i> of road course and road user behaviour by a recognizable road design	Road environment and road user behaviour that support road user expectations through consistency and continuity in road design
<i>State awareness</i> by the road user	Ability to assess one's task capability to handle the driving task

3.2. Dealing with physical vulnerability: homogeneity and forgivingness principles

If road users perform dangerous actions that lead to crashes, the human body's integrity is jeopardised. This vulnerability results from the release of kinetic energy and the body's biomechanical properties.

To deal with the issue of vulnerability in a proactive fashion, sustainable safety requires that controls are placed on factors that may intensify the severity of a crash: differences in speed, direction and mass. This forms the foundation of the *homogeneity principle*. This principle states that where vehicles or road users with great differences in mass have to use the same road space, speeds will have to be so low that, should a crash take place, the most vulnerable road users involved should not sustain fatal injuries. In addition, where traffic is moving at high speeds, road users should be separated spatially. Based on both crash tests between pedestrians and cars, and on ideas developed in the Swedish Zero Vision (Tingvall and Haworth, 1999), the updated sustainable safety vision proposes safe speeds for different situations as shown in Table 2.

Unfortunately, we do not yet have sufficient scientific knowledge to define safe speeds for motorized two-wheelers and heavy vehicles. This issue has also not yet been resolved in practical terms. Separation from other traffic would be the best solution, but it is unclear how this can be realised in practice.

The principle of *forgivingness* (a forgiving roadside) can contribute to reducing injury severity in crashes. Apart from the physical aspects of forgivingness, this principle also has social implications which are more relevant for preventing dangerous actions, as will be discussed in the next sections.

3.3. Prevention dangerous actions: predictability and state awareness principles

People can perform tasks at different levels of control: skill-based, rule-based or knowledge-based (Rasmussen, 1983). Generally speaking, the longer people are trained in performing a task, the more automatic their behaviour. The benefit is that task execution takes less time, effort and attention, and that fewer (serious) errors are made (Reason, 1990). To prevent dangerous actions as much as possible, sustainable safety strives to avoid knowledge-based task performance in particular. One of the conditions for this is that people have to be sufficiently capable (trained) and experienced to take part in traffic. However, the environment can also contribute to this by supporting expectations of road users about their own appropriate behaviour and what they can expect from other road users. This is manifest in the *predictability principle*, the benefits of which can be delivered, according to the updated sustainable safety vision, by *consistency and continuity* in road design (see also Krammes et al., 1995; Lamm et al., 1999). This means that the design needs to support the user's expectations of the road, and that all components of the design need to be in line with these expectations. Ideally, the picture of the road environment is so clear for road users that it can be considered to be "self explanatory" (Theeuwes and Godthelp, 1993). In such a case, the road user needs no additional information to use the road safely. Conversely, if the road environment meets user expectations insufficiently, road users may miss relevant objects and delay the action needed to prevent a crash (see e.g. Theeuwes, 1991; Theeuwes and Hagezieker, 1993).

People not only act dangerously because they make errors unintentionally, they can also exhibit dangerous behaviour by intentionally violating traffic rules. The original sustainable safety vision did not yet take these "unwilling" people into account, but the updated version includes them. In situations where the road environment does not stimulate proper behaviour, a sustainably safe road traffic system benefits from road users who spontaneously obey traffic rules from a normative point of view. To achieve this, traffic regulations have to match the environment, and people have to be educated about the logic and usefulness of rules (Kelman,

Table 2

Proposal for safe speeds in particular conflict situations between traffic participants (Tingvall and Haworth, 1999)

Road types combined with allowed road users	Safe speed (km/h)
Roads with possible conflicts between cars and unprotected road users	30
Intersections with possible conflicts at right angles between cars	50
Roads with possible frontal conflicts between cars	70
Roads with no possible frontal or lateral conflicts between road users	≥ 100

2001; Yagil, 2004). Where people still fail to comply with the rules, policing to a level where a reasonable chance of being caught is perceived is the usual measure to enforce compliance.

Another element in the updated vision is that traffic has to be sustainably safe for *everybody*, and not just for “the average road user”. Fuller’s task-capability interface model (Fuller, 2005) supplies a theoretical framework here. Fuller’s model states that road users’ task capability is the sum of their capacities less the sum of their impairments caused by their present condition (e.g. because of fatigue or use of alcohol). For safe road use, the task capability has to be large enough to meet the task requirements. These task requirements are primarily dictated by the environment, but they can also be altered by the road user, for instance by increasing or decreasing driving speed.

A new element in sustainable safety is the principle of *state awareness*. This principle requires that road users should be able to assess their own task capability for participating in traffic. Task capability can be insufficient due to a lack of competence (e.g. because of a lack of driving experience), or because of – or aggravated by – a state of mind that temporarily reduces the task capability (e.g. because of fatigue, or the use of alcohol or drugs). Since task capability differs between individuals (e.g. inexperienced and elderly road users with underdeveloped or diminished competencies, respectively and also fatigued “average” road users or road users under the influence of alcohol or drugs), generic road safety measures are a necessity for safe traffic. However, for the group of road users with a lower task capability in particular, these measures are not sufficient for safe participation in traffic. Therefore, *generic measures* have to be supplemented with *specific measures* aimed at these groups or situations involving them. Specific measures can be found in the areas such as regulation, education, enforcement (e.g. banning drivers under the influence of alcohol or drugs), and Intelligent Transport Systems (ITS).

Dangerous actions can also be affected by explaining and gaining support for the principle of forgivingness. More experienced road users can, by means of forgiving driving behaviour (in terms of being anticipative or defensive), increase the room for manoeuvre of less experienced road users. Errors should still be regarded as errors by the less experienced in order that they can learn but a forgiving approach should lead to fewer or less serious crashes.

4. Sustainable safety to date: effects and lessons

Sustainable safety has caught on in The Netherlands and it has become a leading vision to further improve road safety. It is apparent that sustainable safety appeals to, and is valued by road safety professionals and is internationally regarded as an authoritative vision (Peden et al., 2004).

After the launch of sustainable safety in 1992, several steps were taken to implement road safety measures in line with the vision in the Netherlands. Perhaps the most important step was the *Sustainable Safety Start-Up Programme*: a covenant with 24 agreements between four levels of administration: at national, regional, municipal level and the district water boards (VNG et al., 1997). Making road infrastructure safer was a visible prime consideration in the execution of sustainable safety. This thinking was both understandable and correct (“*crash occurrence is a priori dramatically reduced by infrastructure design*”) (Koornstra et al., 1992). Nevertheless, this narrow interpretation does not do full justice to the vision; the vision was actually broader in orientation. Page 20 reads: “*The sustainably safe traffic system has an infrastructure that is adapted in design to human capabilities, vehicles having means to support and simplify human tasks and that are constructed to protect the vulnerable road user, and a road user who is trained, educated and informed adequately, and controlled where necessary.*” On this basis, the vision certainly has been translated into road infrastructure design and arrangement, adapted to human capabilities, both in terms of road design handbooks and guidelines and in actual road design (Wegman et al., 2006).

Improvements in vehicle passive safety have been advancing, e.g. through EuroNCAP (European New Car Assessment Programme). Although these developments are at the heart of sustainable safety, they do not appear to have been stimulated by sustainable safety. The same is the case for in-car and out-of-car provisions to simplify and assist driver tasks, particularly in the area of ITS (Intelligent Transport Systems).

Finally, with reference to the: “*road user who is trained, educated and informed adequately, and controlled where necessary*”, we have to conclude that the sustainable safety perspective has not been very inspiring in

realising this ambition. The three areas of driver/rider training, traffic education and police enforcement have advanced, but relatively independently of sustainable safety.

Does the *Sustainable Safety Start-Up Programme* give a good synthesis of the sustainable safety vision? In broad terms it does, provided we accept that the objective was to implement measures relatively quickly. For instance, the basic agreement concerning the categorization of roads has been of great importance. Putting access roads to the fore has been a conscious choice within the *Start-Up Programme*. This was an attractive idea because there was much support within the population in general to do something about the problems on this type of road. It also created the opportunity to categorise the whole road network, which has now been completed. However, the emphasis on access roads has drawn attention away from distributor roads, which have a comparatively high crash risk. Despite the fact that this was understandable and reasonable (the problems are great and the possibilities for solutions limited), this meant that a large part of the problem has not yet been tackled, apart from the construction of roundabouts.

With respect to accompanying policy, the *Start-Up Programme* has greatly facilitated the dissemination and sharing of acquired knowledge, particularly between local authorities. Websites, brochures, newsletters, platforms and working groups provided ample evidence of this. The Infopoint Sustainable Safety has played a central role here. However, one of the points that was missing was a structural evaluation of measures on which the continuation of sustainable safety could build. The lack of knowledge of education is also worth noting. Much knowledge can still be gained concerning infrastructural measures. This knowledge is necessary to be able to make cost-effective advances in the battle for road safety. Based on the existing knowledge it has been estimated that the aggregate effect of all implemented infrastructural measures within the framework of sustainable safety is a reduction of 9.7% in the number of road fatalities and 4.1% in hospital admissions (Wegman et al., 2006). These reductions were not found as a result of formal evaluation studies, but by combining safety effects in different pilot-studies (e.g. effects of roundabouts, 30 km/h-zones, etc.) with information on the size and nature of sustainable safety measures which were implemented. It is good to keep in mind that investment in sustainable safety is based on cost-benefit analyses, and that the results of earlier calculations are recognized as robust by the Netherlands Bureau for Economic Policy Analysis (CPB et al., 2002).

Advancing with sustainable safety for the coming 15–20 years requires means and methods of implementing this vision within current administrative arrangements. It is recommended to:

- *Create a shift in emphasis.* In the future, there is a clear need to strive for a broader approach. Co-ordination can be achieved by thinking in terms of networks, as proposed in the *Mobility Policy Document*. Road safety considerations should be integrated with those of flow/access and the environment to arrive at rational and transparent choices. Specific knowledge of these subjects is essential to underpinning these choices. A broader approach is also required to further integration of technology and vehicles and elements such as education and enforcement. We recommend developing much fuller integration of measures in the field of infrastructure, vehicle technology, education, and enforcement.
- *Try to prevent diluting the potential safety effect.* We have concluded that in various instances too many compromises were made during the transfer from vision to implementation. From the evaluations of the *Start-Up Programme*, we can conclude that low-cost implementation indeed meant too much dilution. The tendency to implement measures on a low-cost basis should be re-evaluated.
- *Pay more attention to knowledge and knowledge-management.* Knowledge about the effects of sustainable safety measures have been gathered haphazardly, rather than in a structured way. In order to know which measures really improve road safety and which merit (substantial) investment, more evaluation of measures is required.

5. Advancing sustainable safety in brief

The sustainable safety vision has been translated into numerous ideas for practical proposals (Wegman and Aarts, 2005, 2006). The following summary may illustrate this.

5.1. Infrastructure

Infrastructure planning and design is an important subject in sustainable safety. The principles of *functionality*, *homogeneity* and *predictability* (see Table 1) have always been central. We propose to maintain these three principles in the future, with *forgiveness* (a forgiving road environment) added as a fourth principle concerning infrastructure.

Large advances have been made in the translation of the original three principles into guidelines for road design and into practical implementation, showing positive safety results. At the same time, we have to conclude that some problems are still waiting for a solution. With respect to *functionality*, we need to set requirements for categorization plans at the road network level. Furthermore, we will have to keep defining essential characteristics of the three Sustainable Safety road categories, and not to restrict ourselves to the agreed and so-called “essential recognisability characteristics”. We also need to develop essential characteristics for intersections.

The principle of *homogeneity* has been developed further with the idea that prior to a collision, speeds are limited to a level such that only “safe crash speeds” pertain. This idea is not found in existing design guidelines. On distributor roads and access roads outside urban areas, discrepancies exist between these accentuated speed requirements and current practice. Many road authorities struggle to decide how to design and construct these roads in a truly sustainably safe way. Our understanding of *recognisability* and *predictability* of road course and other road users’ behaviour has grown, but not yet to the extent that this principle is put into practice based on our knowledge. The new principle (*forgiveness*) was in fact already embedded in sustainable safety but it is appropriate to position it explicitly. Meanwhile, sufficient knowledge has been gathered to apply this principle in full.

5.2. Vehicles

In the sustainable safety vision, vehicle safety occupies an important position because the outcome of certain crash types is determined by crash speed and direction, and by the protection that the vehicle provides (to the occupants and to crash partners). From this perspective (the perspective of the *homogeneity* principle), stricter requirements need to be put on road infrastructure design (and the related speed limits) and heavy vehicles on the one hand, and on cars relative to vulnerable road users (pedestrians, cyclists, and also motorized two-wheeled vehicles) on the other hand. Travel speeds need to be adapted appropriately. This will have to be the norm for the design of our road traffic.

In the area of primary safety (crash prevention), the development of intelligent vehicle systems comes to mind. In secondary safety (injury prevention), it is to be expected that the process initiated with EuroNCAP will continue to bear fruit in the future. It is advisable to expand crash test types (rear-end collisions – to prevent whiplash, and to promote crash compatibility, and also testing of primary safety). One point of concern is the increasing incompatibility between passenger cars (particularly because of the SUV).

5.3. Intelligent transport systems

The application of Intelligent Transport Systems (ITS) deserves a prominent place in the advanced sustainable safety vision. ITS are an important means of making road safety less dependent on the individual choices of road users. It is estimated (OECD, 2003) that safety-oriented ITS may lead to 40% fewer fatalities and injuries. However, in reality, a substantial part of the possibilities to reach this estimate have not yet fully matured, and it is possible that large-scale implementation may run into a variety of problems. We recommend adhering to strong, promising ITS developments, for areas such as congestion reduction and comfort improvement. Road safety should better be integrated in the development process.

In the area of road safety, attention should for the time being be directed at providing information to road users and at developing warning system variants aimed at speed adaptation and dynamic speed limits (intelligent speed assistant as a support system for road *recognisability*). A second area is to guide road users along the shortest and safest routes, using navigation systems. In a next phase, we can think of more advanced systems, such as ITS applications that control traffic access (valid driving licences key, alcohol interlocks).

Seat-belt interlocks are another possibility. In the still longer term, we will have to think more about automated traffic flow management in order to realise a truly sustainably safe traffic system. Nevertheless, it is worth remarking that it would be unwise to stop applying traditional measures and to wait for the introduction of ITS applications; the future is too uncertain for that.

5.4. Education

In our view, education should aim at five behavioural themes: (1) creating an adequate understanding of the road safety problem and an acceptance of sustainable safety measures as a means to improve road safety; (2) encouraging the making of conscious strategic choices (modal or vehicle choice, route choice); (3) counteracting intentional violations; (4) preventing the development of undesirable or incorrect behaviour; (5) preparing “novices” as much as possible. Education is not a panacea, it cannot be a substitute for other interventions (a sustainably safe road user environment), but it does provide an essential complement to them.

Formal education is required to teach correct behavioural routines; however, practicing these routines needs to take place in informal education. Education’s key task is to focus on those subjects that are difficult to be learned directly from traffic because the relationships cannot be clearly deduced (e.g. the relationship between speeds, crash risk and severity, organization of the transport system, road design and allowed manoeuvres (e.g. understanding the “essential recognisability characteristics”), the danger of overestimation of one’s capacities, and so on). This may also help to make the principles of *self-awareness* and *forgiving road user behaviour* more tangible. More attention needs to be devoted to education aimed at minimizing exposure to dangerous situations.

It is necessary to broaden educational care. To create such a “wider learning environment” consisting of both formal and informal education, co-ordination between organizations and guidance on content are needed in order to help these organizations carry out their tasks competently and with sufficient resource. Central government has an important directorial role to play here.

5.5. Regulations and their enforcement

In sustainably safe road traffic, regulation forms a foundation for the safety management of traffic processes, minimising latent system errors, and restraining risk factors. Ideally, in sustainably safe road traffic people comply with the rules (spontaneously) without having to make an effort and without feeling negative about it. On the one hand, this can be accomplished by adapting the traffic environment (such as infrastructure and vehicles) in such a way that it supports the (prevailing) rules as much as possible. This would be the basis to prevent latent errors in the traffic system, because it tackles the cause of traffic violations at the earliest possible stage. On the other hand, intrinsic motivation could prompt people to comply with rules spontaneously.

Unfortunately, spontaneous traffic rule compliance is far from being a reality and it is highly doubtful that it could be relied on in the future. Not everyone is always motivated to comply with the rules, not even when the environment has been adapted optimally. The threat of penalties is needed to encourage these road users to comply with the rules, for instance by making the cost for non-compliance outweigh the perceived benefits of it. Specific enforcement focused on target groups and inspection prior to taking part in traffic, fits within sustainably safe road traffic (an aid in the principle of *state awareness*). In order to lower the number of violations substantially, intelligent transport systems provide some solutions for the future. To prevent people violating rules unintentionally, intelligent systems can be employed as advisory systems. For dedicated target groups this type of system can also be used as a radical, coercive variant (such as for recidivists or serious offenders).

5.6. Speed management

Speed and speed management are key elements in sustainable safety and setting safe speed limits should be a point of departure for the whole of the Dutch road network. However, many current speed limits are being

very widely ignored, and it has to be acknowledged that some individual road users experience “going fast” as fun, exciting and challenging. SWOV estimated that if everyone were to comply with existing speed limits this would lead to a reduction of 25–30% in the number of casualties (Oei, 2001). If safe speed limits were to be introduced and if road users complied with them the benefits could be even greater. Speed limits need to be credible for the road user; that is: they have to be seen as logical in the given circumstances. In the short term, apart from setting safe and credible limits, good information needs to be given to road users (principle of *predictability*). Furthermore, we have two instruments that have proved effective in the past and that, if put into practice appropriately, will also be usable in the future: physical speed-limiting (physical) measures and police enforcement. In the longer term and making use of ITS, we recommend that speed limits to be made dynamic. This will result in speed limits that are not coupled inflexibly with a given road but are adjusted to prevailing conditions.

In order to attain sustainably safe speeds in The Netherlands, the phased plan below can be used:

- to identify criteria for safe and credible speed limits and minimum requirements for road user information;
- to survey the Dutch road network in order to assess if the road environment and the existing speed limits are in conformity with each other, and to implement adaptations (to the road environment or the speed limit) where necessary;
- to re-orientate regarding enforcement of speeds of intentional violators;
- to prepare for and to introduce dynamic speed limits.

5.7. *Drink and drug driving*

The approach to combating driving under the influence takes place at several levels: through legislation, police enforcement, education, punishment, rehabilitation and exclusion. In some of these areas, considerable further gains can be achieved. Police enforcement on alcohol use has doubled since 2000, particularly since the setting up of dedicated traffic police enforcement teams. More than two million road users are tested for alcohol annually. We recommend the dedication of part of the total enforcement capacity to serious offenders. Much can also be improved in the area of rehabilitation of alcohol offenders, particularly by fitting the cars of more serious offenders with alcohol locks (principle of *state awareness*).

Of all new measures mentioned against driving under the influence of alcohol, the alcohol interlock (or alcolock) fits best with the sustainable safety vision. The alcohol interlock has proved effective with convicted drivers and the system should be introduced in The Netherlands as quickly as possible. Perhaps in the longer term all cars can be fitted with alcohol interlocks. However, before that decision is taken, the question of whether or not compulsory use of alcohol locks for all road users yields a safety benefit that outweighs the costs and other possible disadvantages must be answered. Lowest possible limits (so-called zero limits) need to be established for all drugs used in combination with other drugs or with alcohol.

5.8. *Young and novice driving*

Generally speaking, the start of a driving or riding career corresponds with a relatively high risk of crash. The comparatively high risks are caused by a combination of lack of experience and age-specific (biological, social and psychological) characteristics. A sustainably safe environment will lead to lower risks because the lack of experience is compensated for by a safer environment (generic measures). This risk can be reduced further by ensuring that young people take part in traffic in less dangerous circumstances (specific measures, e.g. driving without passengers at night).

Education and traffic enforcement can be made more effective more easily if the environment has been designed to be sustainably safe. Less emphasis should be put on education and driver or rider training in basic skills, and more on acquiring an understanding of the traffic system and of their own capacities. Formal and informal learning should reinforce each other. The graduated driving licence for novice drivers is an effective approach.

5.9. Cyclists and pedestrians

Pedestrians and cyclists are vulnerable in crashes with other types of road users because they are unprotected and also because other types of road users move at higher (sometimes too high) speeds. Crash speeds of motorized vehicles need to remain below 30 km/h in order for pedestrians or cyclists to survive the crash. This means that pedestrians and cyclists have to be separated from high-speed traffic. If this is not possible, the result of conflicts should be such that pedestrians or cyclists are not seriously injured (*forgiveness*). This requires both provisions for motorized vehicles (“friendly” car fronts, and under-run protection for heavy goods vehicles and buses) and for a speed reduction for these vehicles.

The downward trend in crash statistics for pedestrians and cyclists show that we are on the right track. So, the slogan could be: proceed on the chosen path. This path comprises: mix traffic where speeds are low, separate traffic where speeds are too high, and introduce targeted speed reduction where pedestrians and cyclists meet motorized traffic flows. Incidentally, it is also logical to address pedestrians and cyclists about their own responsibilities for safe road use; that they behave predictably, for instance using their bicycle lights at night, and do not cross streets while the lights are red. This will also eliminate a cause of crashes.

5.10. Motorized two-wheelers

Only a few sustainable safety measures provide a truly substantial casualty reduction in crashes involving motorized two-wheeled vehicles. This leads us to a fundamental discussion concerning risk acceptance in a risky society, and to the questions of what is a reasonable and responsible expectation of risk reduction, the distribution of individual and collective responsibility with respect to risk-associated behaviour, and so on. We advocate a fundamental discussion on this topic.

We need to facilitate the safest possible way of using motorized two-wheeled vehicles, given their inherently dangerous characteristics. There are definitely some, although limited, possibilities: obstacle-free zones, advanced braking systems, ITS to influence speeds, conspicuity at crossings, and extra enforcement for mopeds and light mopeds. In rider training, much more emphasis needs to be given to recognizing and anticipating dangers.

5.11. Heavy goods vehicles

According to the sustainable safety vision everything possible must be done to prevent unnecessary movement and then to manage the mileage travelled as safely as possible. Learning from other transport modes and based on an analysis of heavy goods vehicles’ safety problems, the following divisions are recommended:

- two designated road networks for heavy goods transport and light goods transport;
- two vehicle types adapted to the road and traffic situation;
- two types of drivers with different skills requirements.

The leading idea is to separate heavy goods vehicles and other traffic in place and time as much as possible. To this end, a logistics system will need to be developed where heavy goods vehicles use the major road network, and are as little as possible in contact with other, mostly more vulnerable, road users. “Light goods vehicles” made compatible with other traffic then use the remaining road network. Furthermore, the logistics system should be designed such that safety is a design requirement, as is common practice in other transport modes. This also means that the sector develops additional professional skills further. It is also important that companies improve their own safety cultures.

6. Implementation

At the beginning of sustainable safety the view of implementation had the following characteristics (Koornstra et al., 1992; Wegman, 2001):

- Sustainable safety is a scientifically founded, integrated approach to the traffic system, aimed at reducing the possibility for road user error. The approach strives, amongst other things, for a functionally established road network, predictable traffic situations and homogeneous traffic behaviour, where subsequent implementation needs to be sustained over many years, leading to the maximum possible reduction of road traffic casualties.
- Sustainable safety requires coordination of different tasks, whereby the degree of freedom of the public organizations involved to deviate from the content of sustainable safety is limited to some extent, and where the necessary funding has to be provided based on rational considerations (often expressed in cost/benefit and cost-effectiveness considerations).

Since the end of the 1980s, Dutch government can be characterised by a distinct trend towards decentralisation. In a variety of policy areas, policy design, development and execution has been delegated to local and regional levels. This also applies to road safety policy. Local and regional authorities can, independently, undertake the implementation of road safety measures and deliver tailor-made solutions for their areas. At the same time, the idea has taken hold that organizations other than local and regional authorities are also important actors in road safety policy. For example, non-governmental organizations and pressure groups, driving schools, business drivers and transport companies also determine what happens in road traffic. The implementation of sustainable safety has, therefore, become much more complex in recent years, and in the hands of local and regional authorities and pressure groups to an increasing extent. We can speak of a network of decision making that runs across society.

6.1. Public administration perspectives on implementation

The network perspective on implementation as a coordination process between mutually dependent stakeholders differs from the first perspective, because it considers policy implementation from the position of executive organizations and target groups. This perspective is partly based on a bottom-up approach of implementation, also called the “open” approach of implementation (Hanf and Scharpf, 1978; O’Toole, 1988). Adherents of this approach strongly argue from the position of decentralised executive organizations and target groups. They emphasise the importance of the autonomy of these stakeholders, while advocating the reinforcement of their position by providing additional resources from central government. The policy network approach is a second source of inspiration for the multi-stakeholder perspective. The network approach emphasises mutual dependency between parties and sectors, as well as the need for cooperation and coordination (Mandell, 1990; Kickert et al., 1997). In the multi-stakeholder perspective, policy implementation fails if:

- rigid objectives and policy programmes leave executive organizations and target groups with insufficient room to adapt policy to specific circumstances and conditions for implementation;
- insufficient resources are made available;
- policy does not line up with the objectives, opportunities and knowledge of executive organizations, policy makers in other sectors, target groups and stakeholders.

This diagnosis leads to an implementation strategy which is diametrically opposed to the strategy used at the beginning of sustainable safety: to keep objectives and programming global, acquire support from other sectors, executive organizations and target groups, and provide them with opportunities (resources, information and freedom in policy) in order to contribute optimally to the fulfilment of this policy, and use their knowledge of specific conditions and practical viewpoints for the improvement of the policy content (Dowding, 1995; Marin and Mayntz, 1991; Marsh and Rhodes, 1992; Kickert et al., 1997).

6.2. Sustainable safety as implementation programme or guiding concept

Given the recent changes in the policy context of sustainable safety to a fragmented and decentralised network, the multi-stakeholder perspective, in particular, offers ways forward for optimising implementation. The new implementation context can be described as having a faceted character (sustainable safety is considered

against other interests and sectors) and a strong tendency towards decentralisation, therefore necessitating coordination between mutually dependent actors. Some consequences are discussed below of the multi-stakeholder perspective for sustainable safety.

6.2.1. *Uniformity*

One of the fundamentals of sustainable safety is that a certain amount of uniformity is required. This seems to be at odds with the concept of decentralisation and the multi-stakeholder perspective. However, decentralised implementation or implementation as a facet of an area-wide approach certainly does not exclude uniformity. In many sectors, uniform standards and decentralised production go hand in hand as in construction engineering, for example. It is, nevertheless, important to use existing knowledge in decentralised authorities and other sectors in the establishment of uniform policy measures. This knowledge is indispensable to adapt the uniform package of measures to specific conditions. This requires measures to be developed in dialogue with local authorities. For sustainable safety this can be done by gaining the commitment of local authorities through the creation of road safety agreements with provinces and/or other municipalities (Wegman, 2004). High-quality requirements need to be put into the management of this interaction in order to assure progress and quality.

Moreover, sustainable safety could as a “strong brand” also fulfil a role as a “sensitising concept”. Apart from being seen as a collection of road safety measures, it can be regarded as a mobilising and motivating idea that induces people to think about road safety. In this way, sustainable safety is considered more as a paradigm or framework for a quality assurance system than an operational implementation programme. It is a management concept that supports authorities in decisions with road safety implications.

6.2.2. *Sector versus facet*

The perspective of implementation as rational programming starts from a situation in which sectoral policy is already established: it is developed in a separate, “vertical policy category”. However, sustainable safety measures are, increasingly, established within the framework of broader traffic and transport policy. Road safety policy is, to a lesser extent, an independent policy category. This presupposes that as well as a sectoral approach, a faceted approach is required: at various levels of administration, interaction with other sectors is essential, and this broadening of scope offers new opportunities. For example, benefits can be gained by coordinating and embedding sustainable safety within urban development and spatial planning. In other words, sustainable safety is less frequently decided upon in its own sector, it often has to defend its goals as a facet of other policies in different sectors. This fits with the perspective of implementation as a coordination process between mutually dependent actors.

Mingling in other policy sectors does not make the implementation of sustainable safety any easier. People need to be involved in the arenas of traffic and transport policy and urban planning, and they have to make a strong case for road safety interests. Moreover, they have to negotiate their case in these arenas. Knowledge from cost-benefit analyses can be of service here. At the same time, back-up is essential from forums such as regional dialogue groups where road safety interests are discussed. However, the change of institutional rules (due to policy decentralisation) simply makes involvement in different policy sectors necessary.

In conclusion, Table 3 shows the characteristics of both perspectives in respect of the implementation of sustainable safety. Given the decentralisation process of recent years, and the consequent increase in mutual dependency between parties in the implementation context, it is necessary to base a vision of implementation for the next phase of sustainable safety on the perspective of implementation as a coordination process in a multi-stakeholder environment (see Table 3).

6.3. *Quality assurance*

In order to attain a sustainably safe traffic system, it is important to counteract latent errors. This can be achieved with the aid of quality assurance. Various considerations and developments lead to the conclusion that this link is necessary for the high-quality delivery of sustainable safety, but which is at present lacking. A good example of a situation where such quality assurance is needed is in offering road users a recognisable and understandable road design that facilitates the predictability of the road course and other road users' behaviour. To this end, road authorities should agree on a certain level of uniformity of road design. This

Table 3

Two visions on the implementation of sustainable safety

Implementation as rational programming	Implementation as coordination process in a multi-stakeholder setting
Sustainable safety is an effective concept that has to be implemented as completely and uniformly as possible	Sustainable safety is not static. It is about realising uniformity and an adequate adaptation in dialogue with executive organizations
Central control is the best guarantee for a complete and uniform implementation	Central control leads to adaptation problems and alienates potential partners, whereas central government failed as an ally in the past
Area-orientated policy and faceted policy are detrimental to uniform and complete implementation	Area-orientated policy and faceted policy offer opportunities for adaptation of sustainable safety at decentralised level and proactive involvement of related policy areas
Success is the extent to which the realised measures comply with the ideal of sustainable safety	Success is comprised of road safety benefits relative to existing situations
Research institutes contribute to the content of sustainable safety based on their scientific knowledge	Knowledge about sustainable safety facilitates regional and local authorities and other actors in the preparation of measures with road safety impacts

possibility does not exist currently but is fully accepted in other branches of the traffic system. For example, transport companies are required by law to incorporate safety into their business (safety assurance systems). However, this is not yet reality within road transport operations, apart from the transport of hazardous materials. The system of (overarching) quality assurance should be an addition to the quality control that each organization concerned provides itself. Quality assurance will have to be directed at all road traffic components. We recommend conducting a survey into this subject.

It is interesting to see how politicians judge the observed “quality deficit” and the desirability of inspection as part of quality assurance for the further implementation of sustainable safety. More research is needed to prepare this political choice, weighting the benefits and disadvantages. If the choice were to install a (central) supervisor, then its involvement would have to be such that the autonomous competences of authorities are not affected, assuming that those responsible keep and fulfil their own responsibilities. That is: one knows the rules, norms, requirements and so on, and one acts accordingly, or requires third parties (e.g. contractors) to act accordingly. This should satisfy the requirement for the first step of quality assurance (competencies and capabilities are sufficiently covered).

The quality assurance system needs to have a legal basis. We recommend developing this system initially for road authorities. Legislation could take the form of a framework law or principle law as a basis for (delegating) arrangements concerning road safety priorities. A phased structure can be chosen in such framework or principle law aimed at road authorities. This could look as follows:

- Restricting unclear commitments by supervision of road authorities at arm’s length; a basis is constituted for requirements concerning dissemination of information and knowledge, safety assurance systems, training, audits and reviews, terms of reference for contracting, etc.
- The assurance that safety is taken on board and weighted in spatial planning and accessibility plans, e.g. by means of impact assessment reports.
- Conformity and uniformity in infrastructure construction, operation and maintenance.
- Compulsory analysis and remedial action in case of crashes and latent errors.
- Compulsory safety monitoring, both in terms of crash statistics and process indicators.

To avoid misunderstanding: the intention is not to accelerate the implementation of sustainable safety by means of the appointment of a supervisor. The intention is to do it better. To this end, agreements must be made within the regular political and administrative arrangements. Quality assurance should not only be embedded within the organization, but embedded more completely through a supervisor.

7. Conclusions

Although The Netherlands is one of the safest countries in the world in the field of road traffic, substantial improvements can still be made. The sustainable safety vision is a successful method to achieve this. However,

in spite of all its successes, it was beginning to show signs of wear and tear. The latest insights have since been processed and incorporated in the advanced version of sustainable safety.

This paper discusses the road safety situation in The Netherlands and the implementation of sustainable safety from the mid nineties of the last century until now. It resulted in a number of concrete measures, including the introduction of 60 km/h zones in rural areas with mixed traffic, the extension of the number and size of 30 km/h zones in residential areas, the construction of roundabouts and physical separation of bicycles/pedestrians from motorized traffic at higher speed roads. There is still much more to do, but research indicates that, so far, the approach resulted in at least 6% less road traffic fatalities and severe injuries (Wegman et al., 2006). Therefore, the introduction of the sustainable safety vision in The Netherlands can be considered successful.

The advanced sustainable safety vision builds on five safety principles that are based on scientific theories from traffic planning, biomechanics and psychology. This paper gave a brief overview of the background of these principles; three of them (functionality, homogeneity and predictability) were already part of the original sustainable safety vision and thus part of the implementation practices until now. Two new principles were added: state awareness of the road user and forgivingness in a physical and social way. As these principles are new, we have to study how they can be implemented.

On a more detailed, measure-level, issues are indicated that deserve further attention in the future: further improvement of the safety quality of infrastructure, more integration with education (with an emphasis on measures such as “informal education” and education directed at better understanding of rules) and enforcement (with emphasis on prevention of offending against rules). High expectancies are expressed of vehicle and ITS developments, which are part of the market system. Working out complex issues in a complex environment demands the creativity and efforts of many organizations having responsibilities in this area. Political will is an indispensable support in this process.

Whereas The Netherlands is a highly motorized and developed country, the ideas of sustainable safety are certainly not limited to this type of country. The basic idea of fallible and physically vulnerable road users is universal and so are the resulting principles of a sustainably safe traffic system. Clearly, the exact elaboration of the general principles in terms of measures, timing, organization, and financing will need to be different for different (types of) countries and we would like to invite road safety experts to make such an elaboration for their countries. The strong argument that there seems to be no more socio-economically beneficial measure in road traffic than a road safety measure may be a great support in political debates. It is hoped that the advanced sustainable safety vision will inspire further improvements of road safety in the coming decades.

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